



FLEET & SHORE PERSONNEL TRAINING

Anchoring & Mooring Equipment Training

The physical equipment of ground tackle and mooring stations — construction, inspection, safe working loads and the hazards of a failure under load.

SOLAS Ch. II-1

Class Rules

Company PMS

Company SMS

What We Will Cover

01

Ground Tackle & Mooring Equipment Overview

What this equipment does, and why its condition matters

02

The Anchor Windlass & Anchor Chain

Components, operation and chain markings

03

Anchor Types & Ground Tackle Components

Stockless anchors, shackles, swivels and the anchor line

04

Mooring Winches, Ropes & Wires

Winch types and the mooring lines they handle

05

Fairleads, Bollards & Mooring Fittings

The deck fittings that guide and secure every line

06

Inspection, Maintenance & Safe Working Loads

Keeping equipment fit for purpose and knowing its limits

07

Mooring Safety, Snap-Back Zones & Failure Scenarios

Working around stored energy, and what happens when equipment fails



SECTION 1

Ground Tackle & Mooring Equipment Overview

The equipment that holds a ship in place, at anchor or alongside — and why its condition is a safety-critical matter.

01

WHY IT MATTERS

Equipment Under Constant, Heavy Load

Ground tackle and mooring equipment routinely handle forces large enough to injure or kill if they fail or are misused.

- Anchoring equipment holds the ship against wind, current and swell using the combined weight and grip of anchor, chain and windlass brake.
- Mooring equipment holds the ship alongside a berth, buoy or another vessel, resisting the same environmental forces through ropes, wires and winches.
- Both systems store enormous energy under load — a taut mooring line or a running anchor chain releases that energy suddenly and violently if it parts or runs away.
- This equipment is used on almost every voyage, by ratings and officers alike, which makes familiarity — and the complacency that can come with it — a real hazard.
- A single item of failed or poorly maintained equipment can put the ship, the berth structure and any nearby personnel at risk in seconds.



Respect the Load

Every wire, rope, chain and fitting in this course is rated for a specific working load. Understanding — and never guessing at — that rating is the foundation of safe anchoring and mooring.

THE TWO SYSTEMS

Ground Tackle vs Mooring Equipment

Though they share some equipment and principles, anchoring and mooring serve distinct purposes.



Ground Tackle

The anchor, chain cable, windlass and associated fittings used to hold the ship at anchor in open water.



Mooring Equipment

Winches, ropes, wires, fairleads and bollards used to secure the ship alongside a berth or to a buoy.



Shared Principles

Both rely on correctly rated, well-maintained equipment operated by trained personnel following a clear plan.



Shared Hazards

Both involve equipment under high tension, moving machinery and the risk of sudden, violent failure.



SECTION 2

The Anchor Windlass & Anchor Chain

02

The machine that raises and lowers the anchor, and the chain that connects the ship to the seabed.

Windlass Components Every Watchkeeper Should Know

The windlass converts machine power into controlled movement of the anchor cable, and must also hold it securely at rest.



Cable Lifter (Gypsy)

The wheel with pockets shaped to grip the chain links, driving the cable in and paying it out under power.



Brake Band

A friction band that holds the cable stationary once the anchor is deployed, independent of the clutch or motor.



Clutch

Engages or disengages the cable lifter from the drive motor, allowing the cable to run free under brake control.



Chain Stopper / Bow Stopper

A separate mechanical device that takes the working load off the windlass once the anchor is set, for the passage.

Safe Windlass Operation

The windlass is powerful machinery working close to the crew — operating it correctly is a core deck department skill.

- Confirm communications with the bridge and a clear understanding of the intended operation before engaging any control.
- Never engage the clutch and release the brake at the same time without full control of the resulting cable speed.
- Keep hands, feet and loose clothing well clear of the cable lifter, chain and brake band at all times during operation.
- Walk back the cable under power for controlled paying-out in poor holding ground, rather than relying on gravity alone.
- Stop and investigate immediately if the cable runs unexpectedly fast, the brake fails to hold, or unusual noise or vibration is felt.



Never Stand in the Bight

Never stand in the direct line of a running cable or within the area it could snap toward if the brake fails — a chain under load moving suddenly can sever a limb or worse.

Anchor Chain Markings

Chain shackles are marked so the bridge team always knows how much cable is out.

Marking	Meaning
Painted shackle links + studs	Each shackle (27.5 m) marked with paint and wire turns on the studs either side of the joining shackle
Number of turns	Indicates the shackle number, counted from the anchor (1st shackle = 1 turn, 2nd = 2 turns, etc.)
White paint on link before/after	Marks the last link of the previous shackle and the first of the next, for quick visual reference
Deck hawse pipe marking	A painted mark on the chain itself showing when a set number of shackles remain in the locker



SECTION 3

Anchor Types & Ground Tackle Components

From the anchor itself to the smallest connecting shackle, every component of the ground tackle has a defined job.

03

Common Anchor Designs on Merchant Ships

Most merchant ships use a stockless anchor design, though the specific pattern varies.



Stockless (Hall / AC-14 type)

The standard merchant-ship anchor; stows flush against the hull in the hawse pipe with no separate stock to hinder recovery.



Stocked (Admiralty) Anchor

An older design with a stock at the crown, giving strong holding power but requiring more complex stowage; rare on modern ships.



High-Holding-Power (HHP) Anchor

Specially shaped flukes give significantly greater holding power for the same weight, used where extra security is required.



Spare Anchor

Carried on larger vessels per class/flag requirements, stowed and ready to replace a lost or damaged bower anchor.

From Anchor to Hawse Pipe

The ground tackle is a chain of individually critical components, each rated to the same duty as the cable itself.

- The anchor shackle connects the anchor shank directly to the first length of chain cable.
- Joining shackles (Kenter or D-type) connect successive lengths (shackles) of chain, matched in size and strength to the cable itself.
- A swivel piece, fitted within the cable, allows the chain to untwist as the ship swings at anchor, preventing kinking and stress build-up.
- The bitter end is secured inside the chain locker by a quick-release mechanism, allowing the entire cable to be slipped in an emergency.
- Every connecting component must match the chain's own grade and diameter — mixing incompatible grades creates a weak link in a system that is only as strong as its weakest part.



The Weakest Link Rule

A ground tackle system is only as strong as its weakest single component — a mismatched or worn shackle can be the point of failure for an otherwise sound chain.



SECTION 4

Mooring Winches, Ropes & Wires

04

The machinery and the lines that hold a ship firmly alongside, and how to choose and handle each correctly.

Mooring Winch Types

Winch design varies by ship type, but every winch must be able to render (pay out) under excessive load.



Warping Drum / Winch

A simple powered drum for heaving in or paying out a line; the most basic mooring winch arrangement.



Split Drum Winch

Combines a storage drum for the line's bulk with a separate warping (working) drum for handling under power.



Automatic Tension Winch (ATT)

Automatically renders or heaves the line to maintain a set tension as tide and draft change — common on tankers.



Anchor-Mooring Combination Winch

Combines windlass and mooring winch functions on smaller vessels where deck space is limited.

CHOOSING THE LINE

Mooring Rope and Wire Types

Each line material offers a different balance of strength, stretch and handling characteristics.

Line Type	Key Characteristic	Typical Use
Steel Wire Rope	High strength, low stretch, abrasion-resistant but heavy and prone to kinking	Head/stern lines, breast lines on larger vessels
Synthetic Fibre (e.g. HMPE, Nylon)	Lightweight, high strength-to-weight, some stretch depending on fibre type	General mooring lines on most modern merchant ships
Natural Fibre (e.g. Manila)	Lower strength, biodegradable, largely replaced by synthetics	Heaving lines, some traditional/small-craft applications
Combination / Spring Lines	Selected for controlled stretch to absorb surge while alongside	Spring lines resisting fore-and-aft movement

Working Around Rope and Wire Under Tension

Most mooring injuries occur not from equipment failure but from crew positioning themselves poorly around a line under load.

- Never stand in direct line with a rope or wire under tension, on either side of a fairlead, bollard or winch drum.
- Keep fingers and hands clear of the drum, fairlead and any point where a line could nip or crush against a fixed fitting.
- Coil and fake down lines neatly so they can run freely without fouling, snagging feet, or trapping a crew member's leg.
- Wear the full PPE specified for mooring stations, including gloves rated for rope handling and safety footwear.
- Report any sign of wear, broken strands, kinks or corrosion on a line or wire before it is put into service, not after.



SECTION 5

Fairleads, Bollards & Mooring Fittings

05

The deck fittings that guide every mooring line from the winch to the shore, and take the strain once it is made fast.

The Fittings That Carry the Load

Each fitting is designed and rated for a specific role in the mooring arrangement.



Fairlead

Guides a mooring line through the ship's side or over the rail, changing its direction while minimising chafe.



Bollard / Bitts

Paired posts on deck around which a mooring line is made fast, sized to the line's expected working load.



Panama Lead / Chock

A closed or semi-closed fitting in the bulwark or rail that a line passes through, common at Panama Canal transits.



Roller Fairlead

Uses rollers to reduce friction and chafe where a line changes direction sharply under load.

Why Fitting Condition Is Not Optional

A mooring line is only as safe as the fittings it passes through and around.

- Sharp edges, corrosion pitting or cracks on a fairlead or chock can cut into a line under tension, weakening it invisibly from the inside.
- A bollard or bitt must be inspected for cracking at its deck weld or base, since this is where cyclic mooring loads concentrate stress.
- Fittings must be kept free of paint build-up, rust scale and debris that could snag a line or hide developing damage.
- Every fitting has a rated safe working load; a line should never be led around or through a fitting rated below the load it will carry.
- A fitting showing deformation, movement in its foundation, or any visible crack must be taken out of service and reported immediately.



The Line Follows the Fitting

A perfectly rated mooring line routed through a damaged or undersized fitting is only as strong as that fitting — inspect the whole path, not just the rope.



SECTION 6

Inspection, Maintenance & Safe Working Loads

Turning 'it worked last time' into a verified fact, and knowing exactly how much load each item can safely bear.

06

ROUTINE INSPECTION

What to Check, and How Often

Anchoring and mooring equipment should be inspected on a defined schedule under the ship's planned maintenance system.

- Inspect chain cable for wear, elongation, corrosion and cracked links, paying particular attention to the area near the anchor and joining shackles.
- Check the windlass brake lining, clutch engagement and hydraulic or electrical drive systems per the maker's maintenance schedule.
- Examine ropes and wires along their full length before and after each use for chafe, broken strands, kinks, flat spots or corrosion.
- Grease and lubricate fairleads, rollers and winch gearing on the schedule specified in the ship's Planned Maintenance System (PMS).
- Record every inspection, defect found and corrective action taken in the PMS, supporting both safety and class survey requirements.

UNDERSTANDING LOAD RATINGS

Three Numbers Every Mooring Team Should Know

These ratings apply to lines, wires and fittings alike, and should never be confused with one another.

MBL

Minimum Breaking Load — the load at which a new, undamaged item is expected to fail

SWL

Safe Working Load — the maximum load the item should be exposed to in normal service, with margin built in

↓ with age

Both figures fall as wear, corrosion and UV/chemical exposure degrade a line or fitting over its service life

RETIREMENT CRITERIA

When to Take a Line Out of Service

Company SMS and class guidance define objective criteria for retiring mooring lines and wires before they fail in use.

Damage Type	Retirement Guidance
Broken strands (fibre rope)	Retire when broken strands exceed the threshold set in the manufacturer's / Company guidance
Wire rope broken wires	Retire once broken wires in a lay length exceed the applicable class or maker's limit
Severe chafe or flat spots	Retire any section showing significant diameter reduction or core exposure
Heat damage / glazing (synthetic)	Retire immediately — melted or glazed fibres have lost a significant, unquantifiable amount of strength



SECTION 7

Mooring Safety, Snap-Back Zones & Failure Scenarios

Understanding stored energy in a line under load, and what to do when equipment does not behave as expected.

07

STORED ENERGY

Understanding the Snap-Back Zone

A mooring line under tension behaves like a stretched spring — and releases that energy just as suddenly if it parts.

- When a rope or wire under tension parts, the stored energy releases instantly, whipping the line back along a path determined by its geometry.
- Synthetic fibre ropes, particularly high-modulus types, can store significant elastic energy and snap back with very high velocity.
- The snap-back zone is the area a parted line could sweep through — it is not limited to a straight line and can extend well beyond the immediate mooring station.
- Snap-back zones must be marked on deck (hatching or signage) and personnel must be trained to recognise and avoid them during mooring operations.
- A part can happen without warning — sound, sudden slack, or unusual winch behaviour are the only cues, and by the time they are noticed it may be too late to react.



Stay Out of the Zone

No task is worth standing in a marked snap-back zone. If a job requires it, stop, reassess the mooring arrangement, and find another way to do it safely.

Working Safely Around Lines Under Load

Discipline at the mooring station is what turns a routine evolution into a safe one.



Do

- Identify and stay clear of every snap-back zone before lines are put under tension
- Use the correct number of turns on bollards and winch drums for controlled surging
- Communicate clearly with the winch operator before and during heaving or paying out
- Stop the operation immediately if a line shows signs of overload, such as smoking or excessive stretch



Don't

- Don't stand in a bight of line, in a snap-back zone, or between a line and a fixed fitting
- Don't exceed a winch's or fitting's rated capacity to 'get the job done faster'
- Don't leave a line under load unattended without someone monitoring tension
- Don't attempt to free a jammed or fouled line while it remains under tension

WHEN EQUIPMENT FAILS

Responding to an Equipment-Failure Scenario

A clear, drilled response limits the consequences when a line parts, a winch fails or a fitting gives way.

Scenario	Immediate Response
A mooring line parts	Clear the area, account for all personnel, assess for injury, report to the bridge/Master immediately
Windlass brake fails to hold	Stop paying out if possible, warn all nearby personnel clear, engage secondary chain stopper if safe to do so
Winch loses power mid-operation	Apply the mechanical brake, secure the line to a bollard by hand-turns if safe, do not leave the drum unattended
Fitting shows sudden deformation	Ease tension immediately, re-route the line to an alternative fitting, tag the damaged fitting out of service

Investigation and Reporting

Every equipment failure, however minor its consequences, is a source of lessons for the whole fleet.

- Report every parted line, failed brake or damaged fitting to the Master and through the ship's SMS without delay, even if no one was injured.
- Preserve the failed component where practicable — a parted line's break point can reveal whether the cause was wear, overload or a manufacturing defect.
- Review the mooring arrangement and line selection for the berth to determine whether the failure reflects a wider pattern.
- Update the PMS record and inspection routine if the investigation finds an inspection gap that allowed the failure to develop undetected.



Every Failure Teaches Something

A single parted line, properly investigated, can prevent the next one — treat every failure as fleet-wide learning, not a one-off inconvenience to move past quickly.



COURSE COMPLETE

Well-maintained equipment, correctly used, is what keeps a routine mooring operation routine.

Company DPA — 24/7 emergency line

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